

Multiple Choice (10 marks)

1. What is the binding energy of the lithium nucleus?
 - (a) 42.1 MeV
 - (b) 34.8 MeV
 - (c) 39.2 MeV
 - (d) 40.5 MeV
 - (e) 38.7 MeV
2. What Mars mission will be landing on May 25 2008 and will dig a trench into (hopefully) ice-rich soil?
 - (a) Mars Science Laboratory
 - (b) Mars Odyssey
 - (c) Mars Atmosphere and Volatile Evolution (MAVEN)
 - (d) Phoenix Mars Lander
 - (e) Mars Reconnaissance Orbiter
3. Determine the focal length f of a glass lens of index 1.5 surrounded by air for which $r_1 = +10$, $r_2 = +9$ a) when the lens thickness is zero, b) when the thickness is 1.
 - (a) a) 180 units, b) 270 units
 - (b) a) -360 units, b) -240 units
 - (c) a) 270 units, b) 180 units
 - (d) a) -180 units, b) 180 units
 - (e) a) -270 units, b) -180 units
4. By definition, the electric displacement current through a surface S is proportional to the
 - (a) rate of change of the magnetic flux through S
 - (b) electric flux through S
 - (c) rate of change of the electric current through S
 - (d) time integral of the magnetic flux through S
 - (e) rate of change of the electric flux through S
5. What is the wavelength of a 10-eV electron?

- (a) 2.5 Å
- (b) 3.86 Å
- (c) 4.5 Å
- (d) 1.5 Å
- (e) 1.24 Å

True / False (10 marks)

Answer True or False

- 6. True or False: In a fusion reaction, the majority of energy released is in the kinetic energy of the alpha particle and neutron equally.
- 7. True or False: The phenomenon of beats is primarily caused by the refraction of sound waves in different media.
- 8. True or False: The focal length of a thin lens remains unchanged when it is immersed in a medium with a lower refractive index than air.
- 9. True or False: The distance from Earth to the star Altair is approximately 9.3×10^{10} meters.
- 10. True or False: The gravitational acceleration experienced by the Space Shuttle at 300 km altitude is approximately 0 m/s^2 .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Calculate the gravitational force acting on a 70 kg person in an accelerating automobile at 4 m/s^2 . Discuss the implications of this force in the context of Newton's laws of motion.
- 12. A dragster completes a quarter-mile race, reaching a speed of 80 m/s. Calculate the acceleration and time taken for the run, and discuss the implications of these values in the context of drag racing physics.

End of Paper